Intelligent Systems Engineering Department
The National School of Artificial Intelligence
3rd Year, Semester 1 — 2025/2026

NETWORKS AND PROTOCOLS

Exercises Sheet 4

Exercise 1

Consider the following IP datagrams:

<table>
<tr><td>100</td><td>101</td><td>111111xx</td><td colspan="2">0000 0000 0101 1010</td></tr>
<tr><td colspan="3">0000 0000 0000 1010</td><td>0x0</td><td>0001 1000 0110</td></tr>
<tr><td colspan="2">0000 0001</td><td>0000 0110</td><td colspan="2">?</td></tr>
<tr><td colspan="5">00001111.11110000.00001111.10001001</td></tr>
<tr><td colspan="5">00001111.11110000.00001111.11111111</td></tr>
<tr><td colspan="5">Data (Datagram D)</td></tr>
</table>

<table>
<tr><td>100</td><td>101</td><td>111111xx</td><td colspan="2">1011 1001 0011 0000</td></tr>
<tr><td colspan="3">0000 0000 0000 1011</td><td>0x0</td><td>0000 0000 0000</td></tr>
<tr><td colspan="2">0000 0001</td><td>0000 0110</td><td colspan="2">?</td></tr>
<tr><td colspan="5">00001111.11110000.00001111.10001001</td></tr>
<tr><td colspan="5">00001111.11110000.00001111.11111111</td></tr>
<tr><td colspan="5">Data (Datagram B)</td></tr>
</table>

<table>
<tr><td>100</td><td>101</td><td>111111xx</td><td colspan="2">0000 0000 0101 0100</td></tr>
<tr><td colspan="3">0000 0000 0000 1010</td><td>0x1</td><td>0000 0000 0000</td></tr>
<tr><td colspan="2">0000 0001</td><td>0000 0110</td><td colspan="2">?</td></tr>
<tr><td colspan="5">00001111.11110000.00001111.10001001</td></tr>
<tr><td colspan="5">00001111.11110000.00001111.11111111</td></tr>
<tr><td colspan="5">Data (Datagram C)</td></tr>
</table>

<table>
<tr><td>100</td><td>101</td><td>111111xx</td><td colspan="2">TL</td></tr>
<tr><td colspan="3">0000 0000 0000 1010</td><td>0x1</td><td>Offset</td></tr>
<tr><td colspan="2">0000 0001</td><td>0000 0110</td><td colspan="2">?</td></tr>
<tr><td colspan="5">00001111.11110000.00001111.10001001</td></tr>
<tr><td colspan="5">00001111.11110000.00001111.11111111</td></tr>
<tr><td colspan="5">Data (Datagram A)</td></tr>
</table>

1. Which datagrams carry fragments?

2. Calculate the data size of datagram A ;

3. If datagram A is a fragment, calculate its displacement;

4. Reassemble the different fragments of the same datagram;

5. Calculate the total size of the original message.

6. Is there any inconsistency in this message? If so, find it.

Exercise 2

Consider a network with an MTU of 150 bytes.

1. Calculate the payload of an IP packet for this network.

2. Deduce the actual size of a fragment in this network.

3. The original datagram is 576 bytes long. Its ID field is 4345. How many datagrams will this packet be fragmented into?

4. What will be the size in bytes of the ”Data” field in the last fragment?

5. What is the offset field value for each fragment?

Exercise 3

Faced with a number of malfunctions in your network, you decide to install an analyzer. This captures and decodes all fields. It provides you with the following information:

- IP datagram version 4, no particular option or service invoked, checksum correct
- source address 194.225.18.5
- destination address 194.225.18.7
- data field length 40 bytes
- datagram ID 28 246D (D for decimal)
- TTL 255
- DF and M bits are set to 0 respectively
- the data field encapsulates a TCP segment (identification 0x06)

Based on the above information, you are asked to reconstruct all the fields of the datagram as trapped by the analyzer (in Hexa).

Exercise 4

Consider the topology in figure 1:

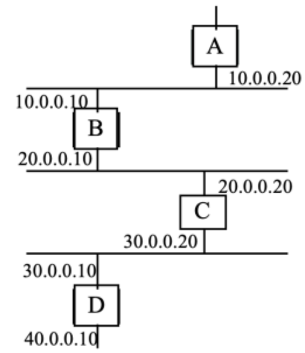

Figure 1: Exercise 4

Subnet 20 only accepts IP packets up to 492 bytes in size. This value is set by the frame format used to transport these packets. The other subnetworks are Ethernet with a maximum frame size of 1518 bytes. Describe what happens at the Ethernet and IP levels, as you pass through Gateway B, when a 1500-byte frame is sent from network 10 to network 30.